COMP 3920 Database Systems 1

MySQL Transaction Predictions Sheet

Introduction:
Use this sheet to predict the outcomes and results of the Transaction Scenarios.

Fill in the predictions BEFORE completing each scenario in this week's lab on Transactions.

Fill in the actual values AFTER completing each scenario in this week's lab on Transactions.

IMPORTANT: Submit this as part of this week's Lab

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Predict the values of the following Transaction Scenarios:

A. Transactions and Rollbacks
B. Transactions and Commits
C. Atomic Transactions
D. Ended Transactions
E. Isolated Consistent Backups
F. MyISAM is Not Transaction Safe
G. MyISAM Read Locks Block Writers
H. Deadlock
I. Implicit Commit with DDL
J. Uncommitted Transactions Prevent Other's DDL
K. Isolation Level – Read Uncommitted
L. Isolation Level – Serializable

Note: Values in the `tx` and `tx_myisam` tables should be reset back (if needed) to the defaults between each scenario.

Scenario A: Transactions and Rollbacks

	Prediction	Actual
What is the value of `value` for `id=1` in Step 5?	1234	1234
What is the value of `value` for `id=1` in Step 6?	1000	1000
What is the value of `value` for `id=1` in Step 8?	1000	1000
What is the value of `value` for `id=1` in Step 9?	1000	1000

Scenario B: Transactions and Commits

What is the value of `value` for `id=1` in Step 5?	Prediction	1557	Actual	1557
What is the value of `value` for `id=1` in Step 6?		1000		1000
What is the value of `value` for `id=1` in Step 8?		1557		1557
What is the value of `value` for `id=1` in Step 9?		1557		1557

Scenario C: Atomic Transactions

What is the value of `value` for `id=2` in Step 4?	Prediction	2000	Actual	2000
What is the value of `value` for `id=3` in Step 4?		3000		3000
What is the value of `value` for `id=2` in Step 9?		2000		2000
What is the value of `value` for `id=3` in Step 9?		3000		3000

Scenario D: Ended Transactions

What is the value of `value` for `id=3` in Step 6?	Prediction	3358	Actual	3358
What is the value of `value` for `id=4` in Step 12?		4000		4000

Scenario E: Isolated Consistent Backup

What is the value of `value` for `id=4` in Step 2?	Prediction	4000	Actual	4994
What is the value of `value` for `id=4` in Step 5?		4000		4994
What is the value of `value` for `id=4` in Step 7?		4994		4994

Scenario F: MyISAM is NOT Transaction Safe

What is the value of `value` for `id=4` in Step 3?	Prediction	40	Actual	40
What is the value of `value` for `id=4` in Step 5?		47		47
What is the value of `value` for `id=4` in Step 7?		47		47

Scenario G: MyISAM Read Locks Block Writers

What happens when you try to run Step 3?	Prediction	The UPDATE hangs/waits, blocked by th
	Actual	The UPDATE hangs/waits, blocked by th
What happens when Step 4 completes?	Prediction	The UPDATE completes immediately aft
	Actual	UPDATE completes immediately after U

What is the value of `value` for `id=4` in Step 5?	Prediction	46	Actual	46
What is the value of `value` for `id=4` in Step 6?		46		46

Scenario H: Deadlock

Which step will create the deadlock?	Prediction	6	Actual	6

Scenario I: Implicit Commit with DDL

What is the value of `value` for `id=3` in Step 3?	Prediction	3518	Actual	3518
What is the value of `value` for `id=3` in Step 4?		3000		3000
What is the value of `value` for `id=3` in Step 6?		3518		3518

Scenario J: Uncommitted Transactions Prevent Other's DDL

What happens when you try to run Step 4?	Prediction	ALTER TABLE hangs/waits, blocked by
	Actual	hangs/waits
What happens when Step 5 completes?	Prediction	ALTER TABLE completes immediately a
	Actual	completes immediately

Scenario K: Isolation Level - Read Uncommitted

What is the value of `value` for `id=2` in Step 3?	Prediction	2000	Actual	2000
What is the value of `value` for `id=2` in Step 5?		2291		2291
What is the value of `value` for `id=2` in Step 7?		2000		2000
What is the value of `value` for `id=2` in Step 8?		2000		2000

Scenario L: Isolation Level - Serializable

What happens when you try to run Step 6?	Prediction	UPDATE completes successfully
	Actual	1 completes successfully
What happens when you try to run Step 7?	Prediction	UPDATE hangs/waits, blocked by SERI
	Actual	2 hangs/waits (blocked by SERIALIZABL
What happens when Step 8 completes?	Prediction	Terminal 2's UPDATE completes immed
	Actual	2 completes immediately after Window #